

PRACTICAL – 3

Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

AIM:

To design responsive layouts for the Home, About, Registration, Dashboard and Events pages of StudentHub using CSS Grid and Flexbox, so that the portal adapts correctly to mobile, tablet and desktop screens, following a mobile-first approach.

Introduction:

The pages built in Practical 2 used a single fixed layout. In this practical, CSS Grid, Flexbox and media queries were added so that the same HTML rearranges itself for different screen widths. The base CSS rules (written without any media query) target mobile screens first, and larger layouts are added afterwards using min-width media queries. No CSS framework such as Bootstrap or Tailwind was used; Grid and Flexbox alone were enough to build the layouts.

Tools and Technology:

VS Code, HTML5, CSS3 (Grid, Flexbox, media queries), Chrome DevTools responsive/device mode, Git and GitHub.

Layout Technique Used on Each Page:

Page	Layout technique	What changes across screen sizes
Home	CSS Grid (.features-grid) and Flexbox (.feature-card, header)	1 column on mobile → 2 on tablet → 3 on desktop; header switches from a stacked block to a single row
About	CSS Grid (.about-layout)	Single column on mobile and tablet; a 2fr / 1fr two-column grid from tablet width up
Registration	CSS Grid (.register-layout, .form-row)	Form and intro text stack in one column below 900px; form fields such as name/email sit two per row above that
Dashboard	CSS Grid (.dashboard-layout, .stats-grid)	Sidebar and content stack in one column on mobile; a 220px sidebar + content grid from tablet width, with a 4-column stats grid on tablet and up
Events	CSS Grid (.events-grid)	Single column of event cards on mobile and tablet, matching the available width at each size

Screenshots:

Each page below is shown at Desktop (about 1588px), Tablet (about 768px) and Mobile (about 375px) width.

Home Page

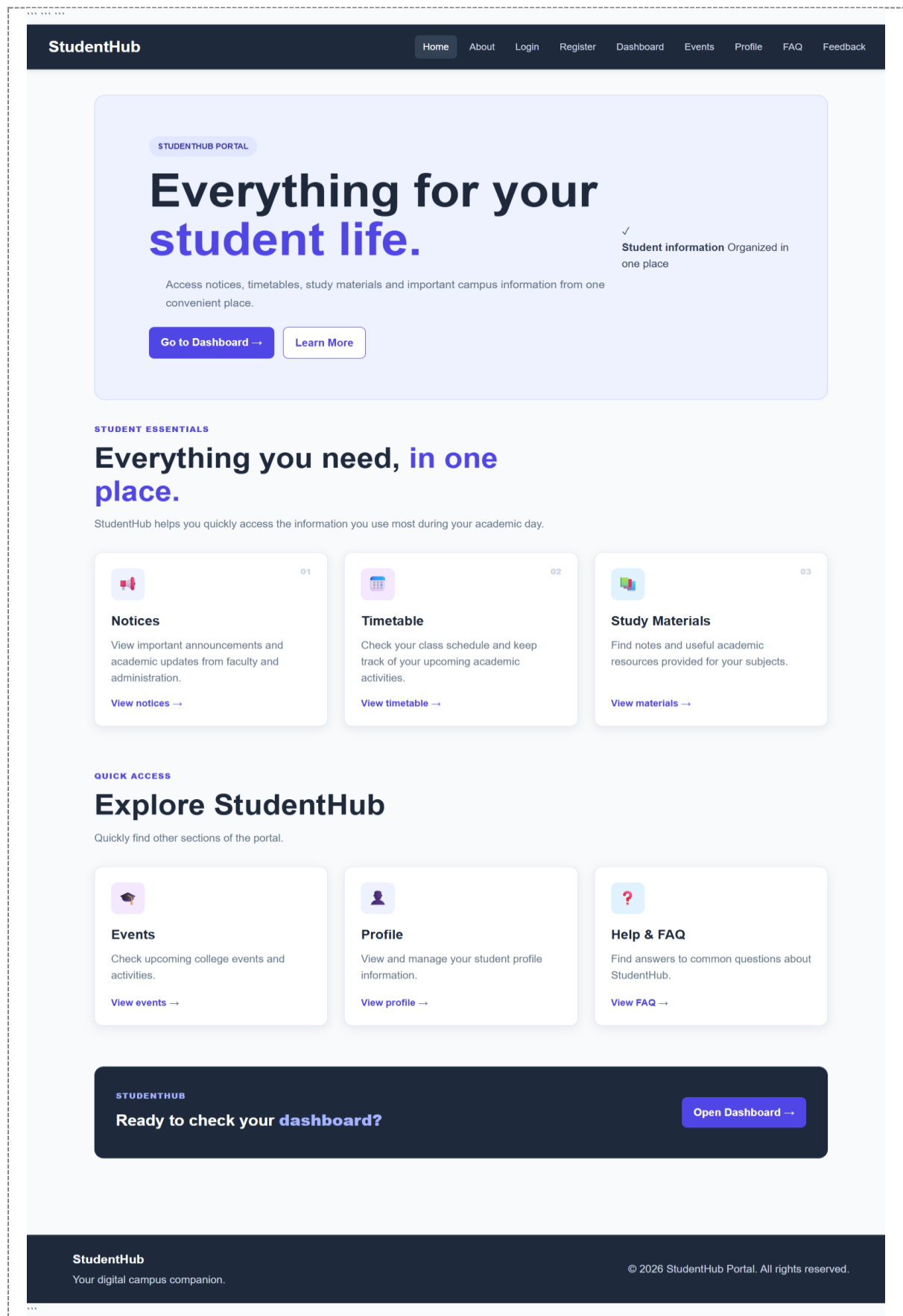


Figure: Home page, desktop width

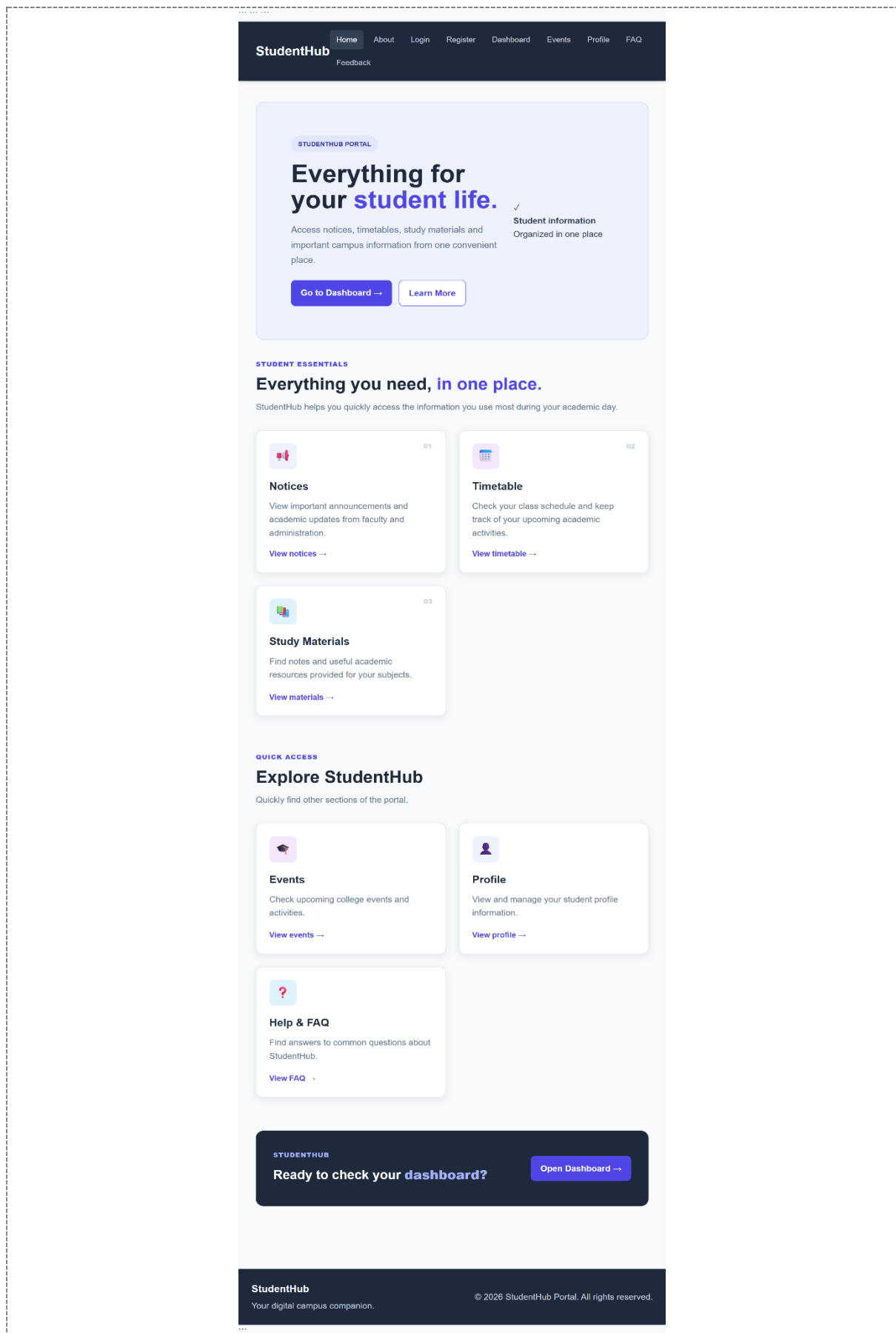


Figure: Home page, tablet width

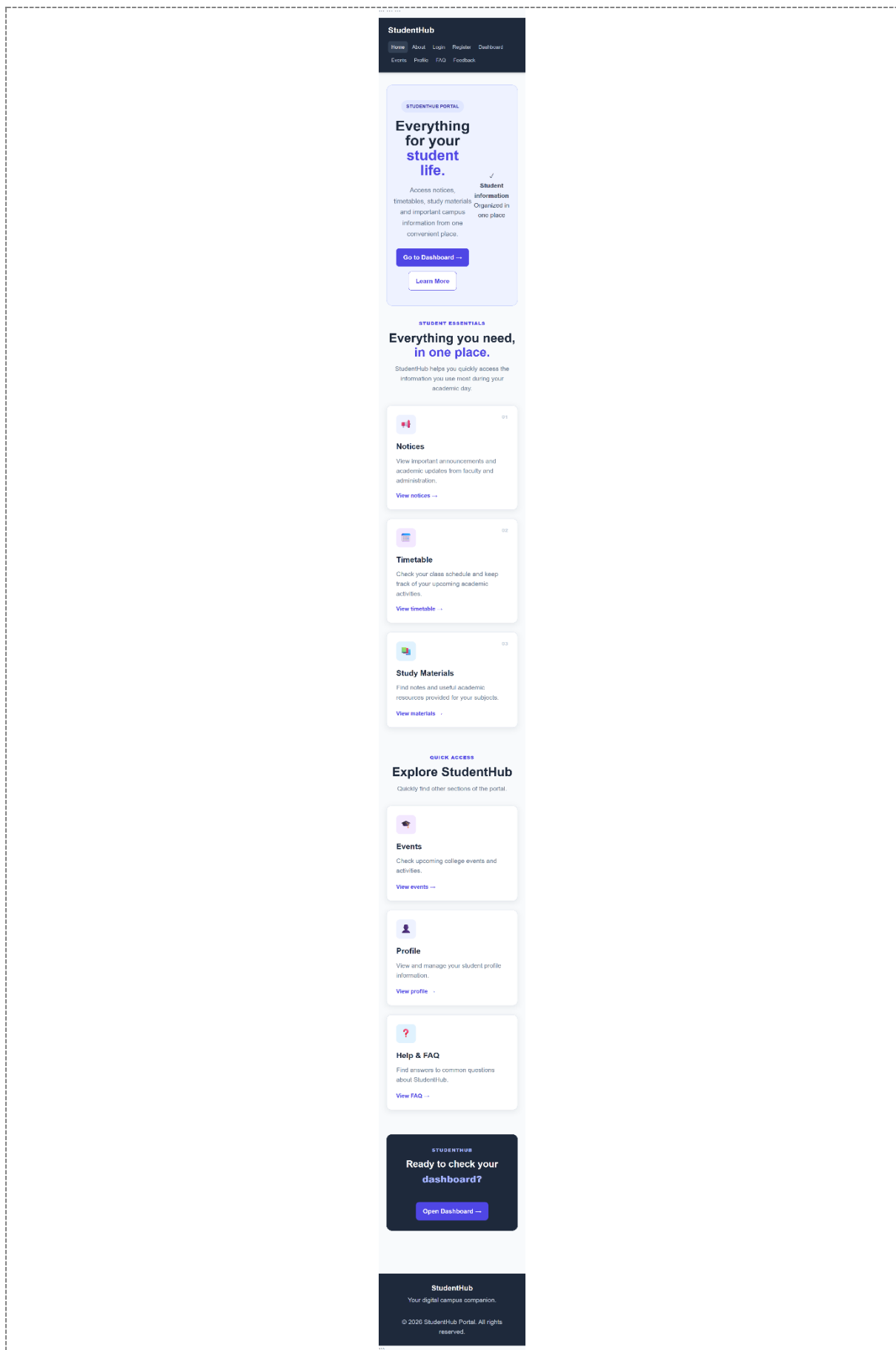


Figure: Home page, mobile width

About Page

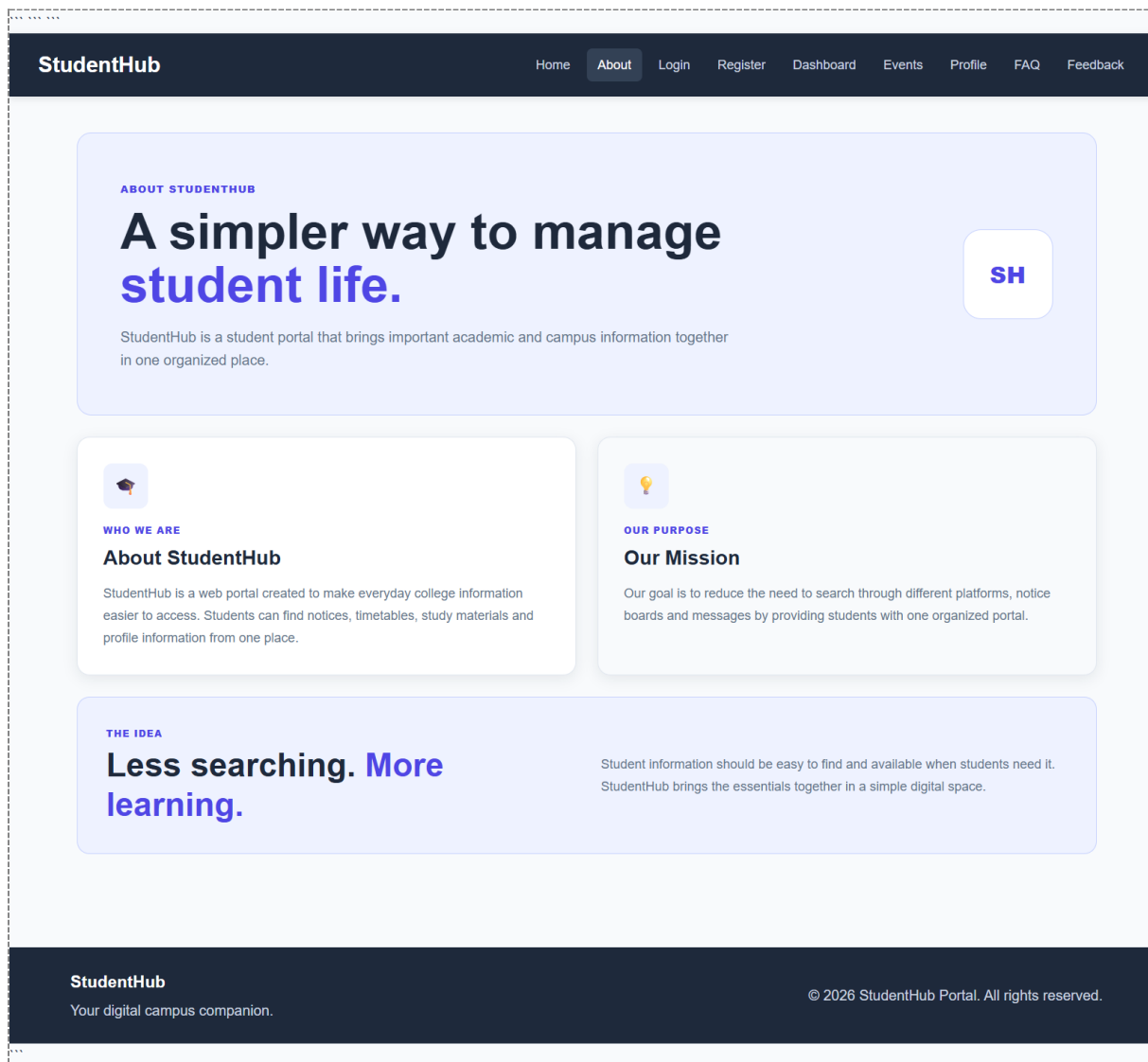


Figure: About page, desktop width

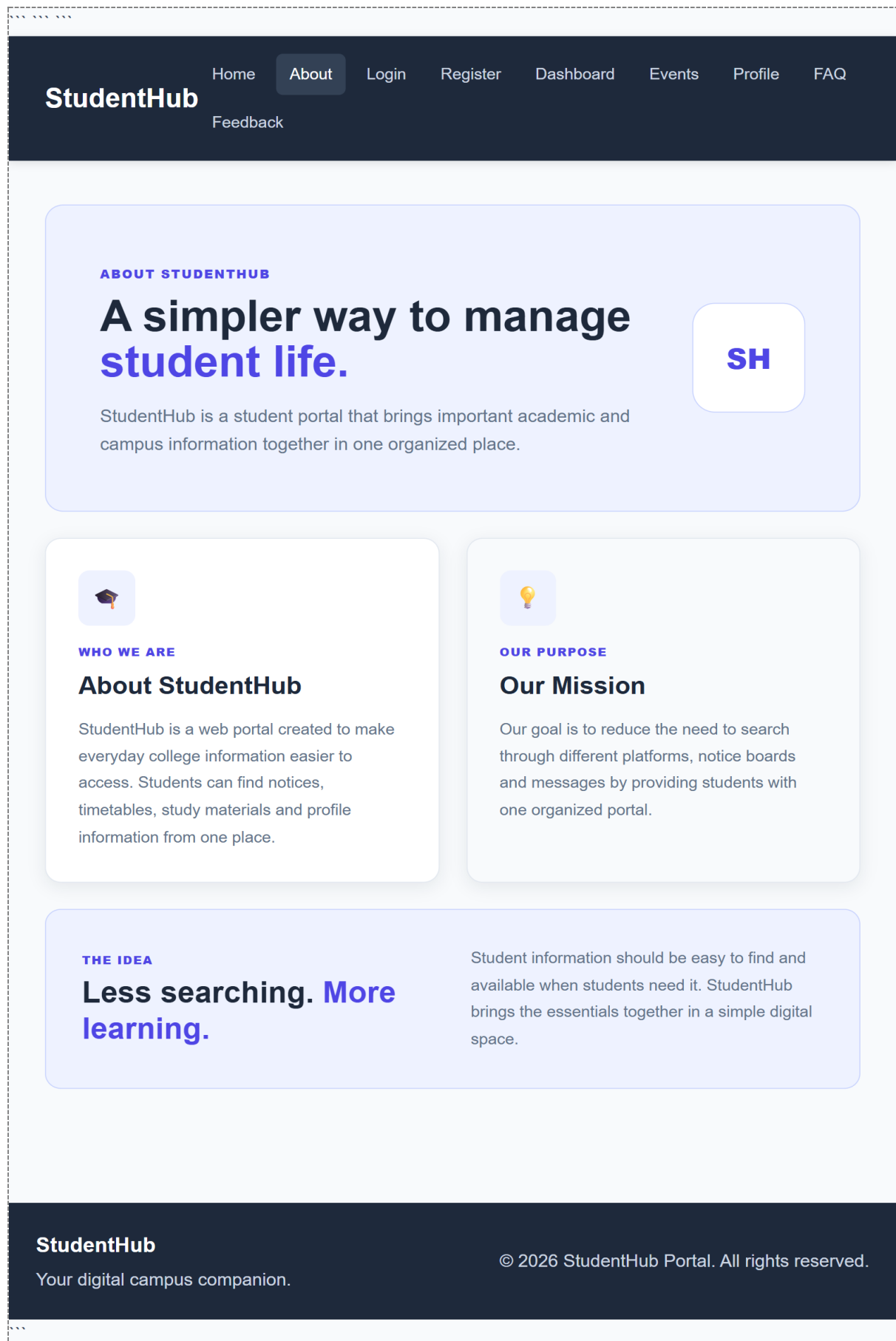


Figure: About page, tablet width

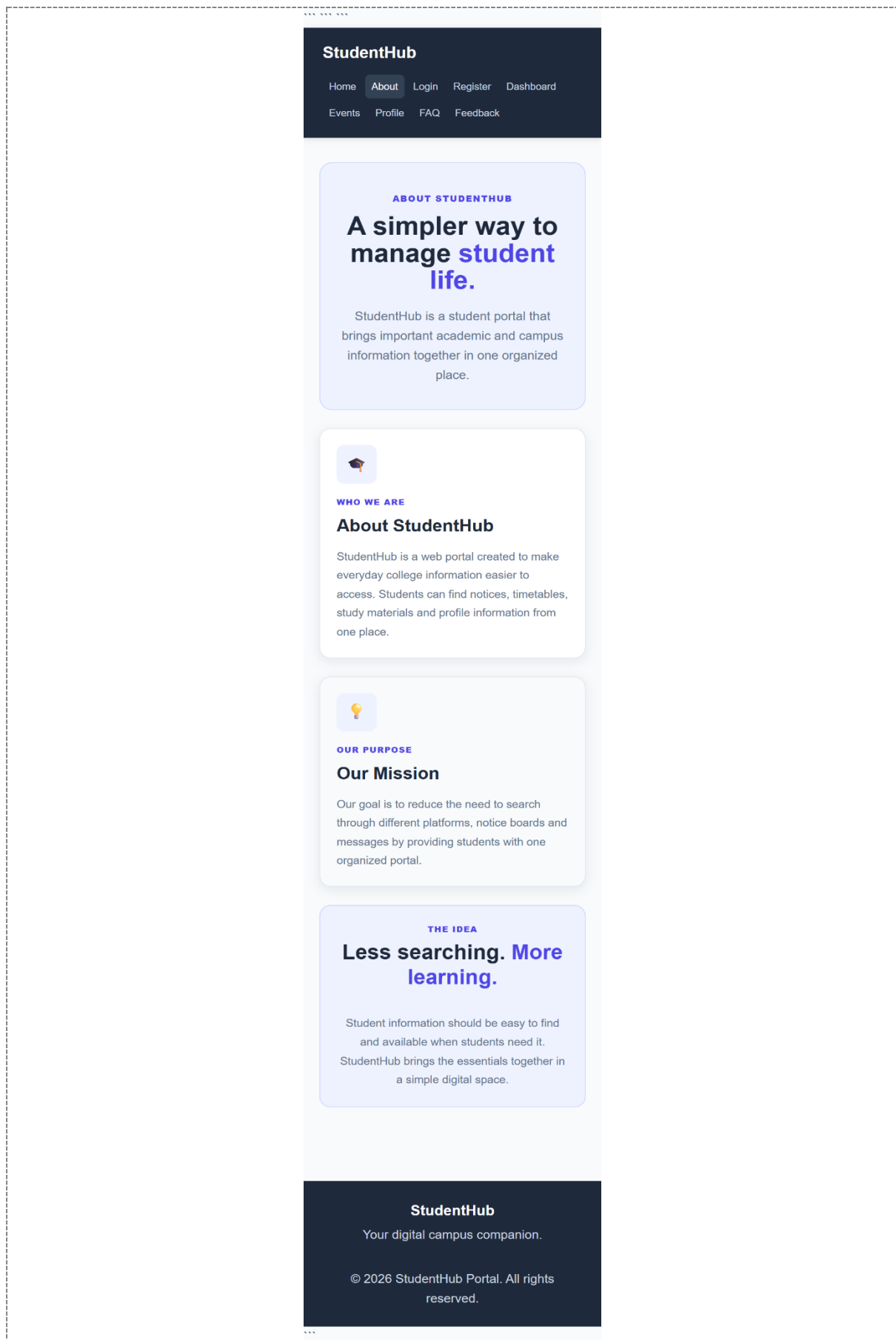


Figure: About page, mobile width

Registration Page

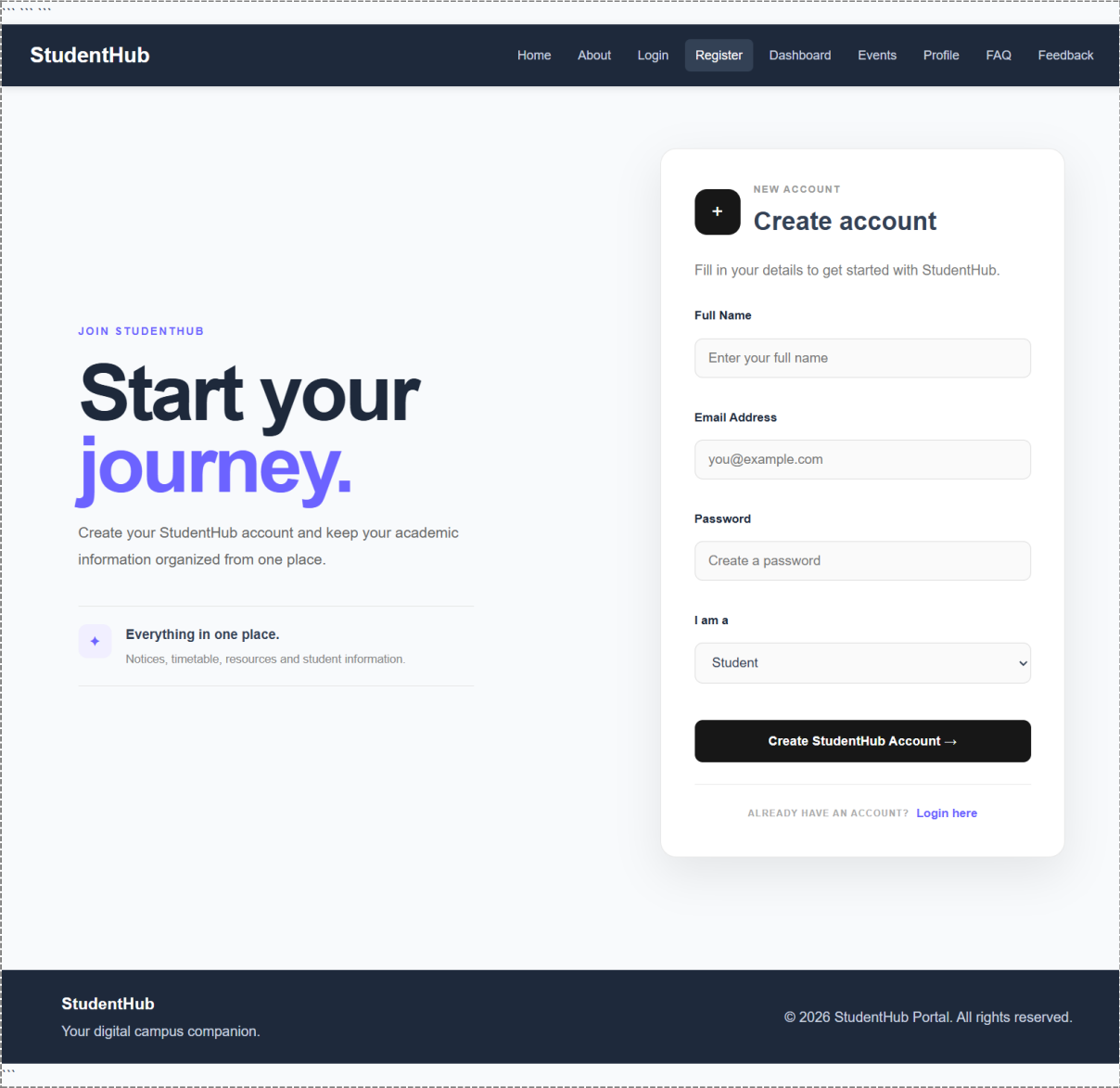


Figure: Registration page, desktop width

StudentHub

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JOIN STUDENTHUB

Start your journey.

Create your StudentHub account and keep your academic information organized from one place.

+

Everything in one place.
Notices, timetable, resources and student information.

NEW ACCOUNT

+

Create account

Fill in your details to get started with StudentHub.

Full Name

Enter your full name

Email Address

you@example.com

Password

Create a password

I am a

Student

Create StudentHub Account →

ALREADY HAVE AN ACCOUNT? [Login here](#)

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Figure: Registration page, tablet width

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JOIN STUDENTHUB

Start your journey.

Create your StudentHub account and keep your academic information organized from one place.

Everything in one place.

Notices, timetable, resources and student information.

NEW ACCOUNT

+

Create account

Fill in your details to get started with StudentHub.

Full Name

Enter your full name

Email Address

you@example.com

Password

Create a password

I am a

Student

Create StudentHub Account →

ALREADY HAVE AN ACCOUNT?

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Figure: Registration page, mobile width

Dashboard Page

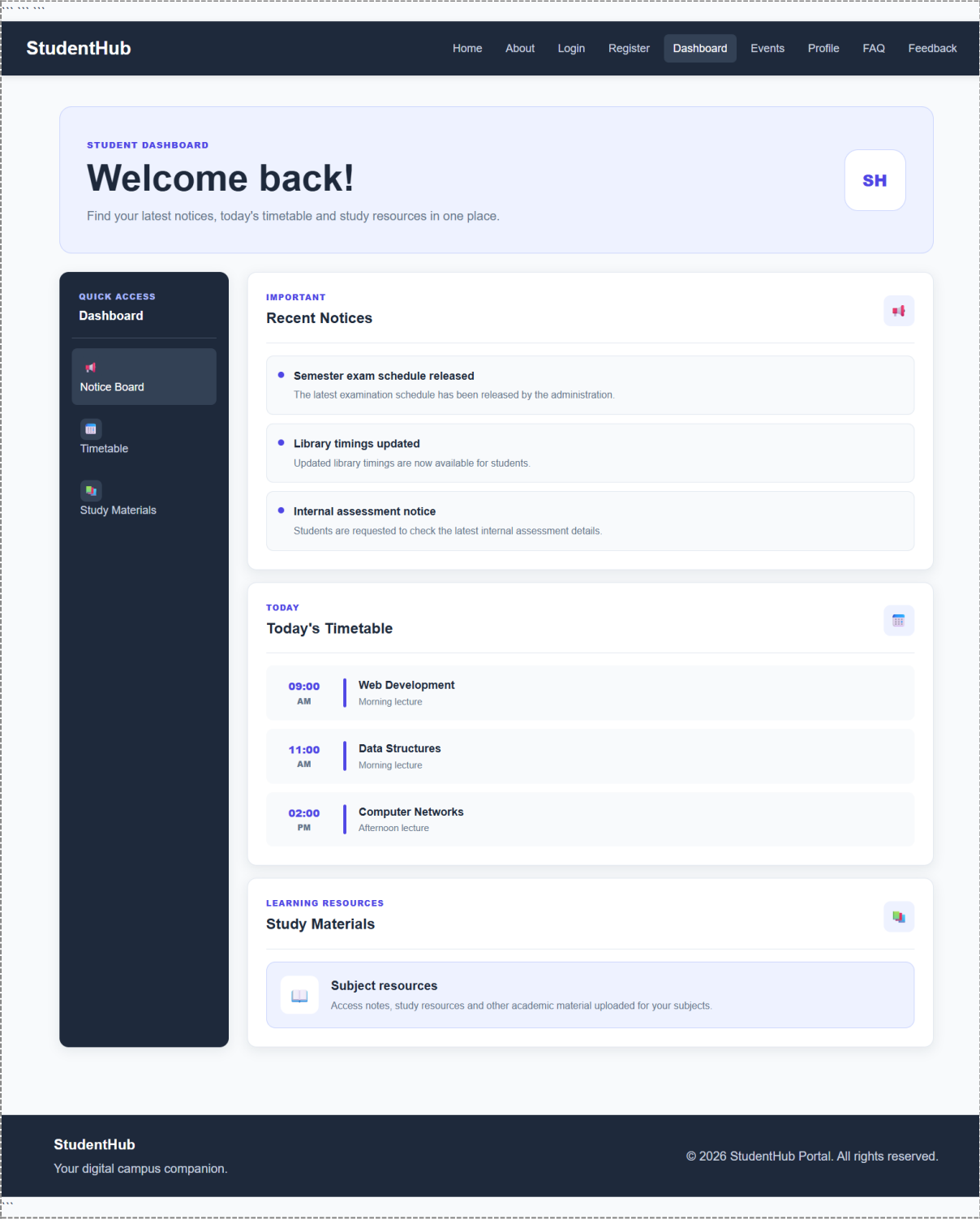


Figure: Dashboard page, desktop width

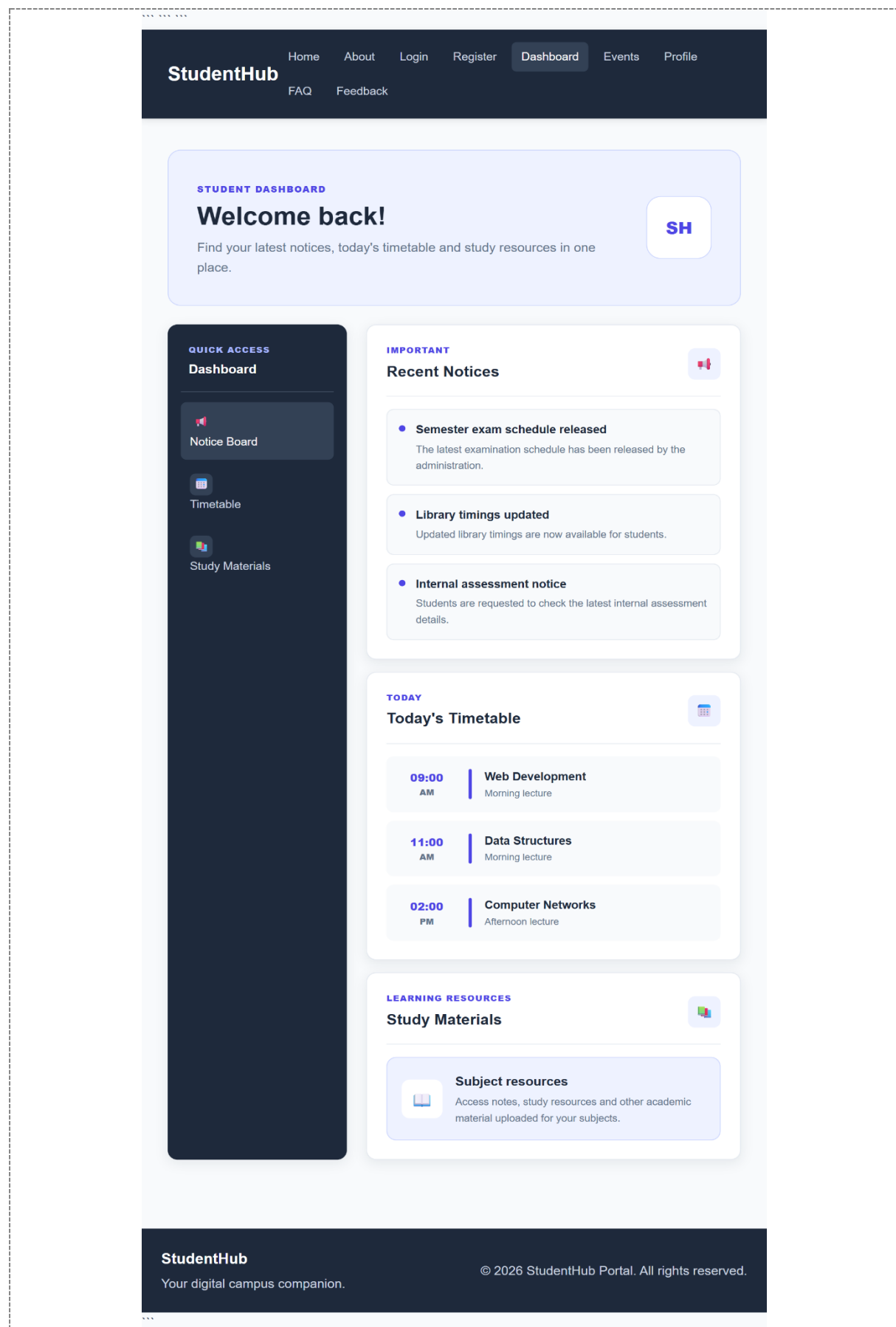


Figure: Dashboard page, tablet width

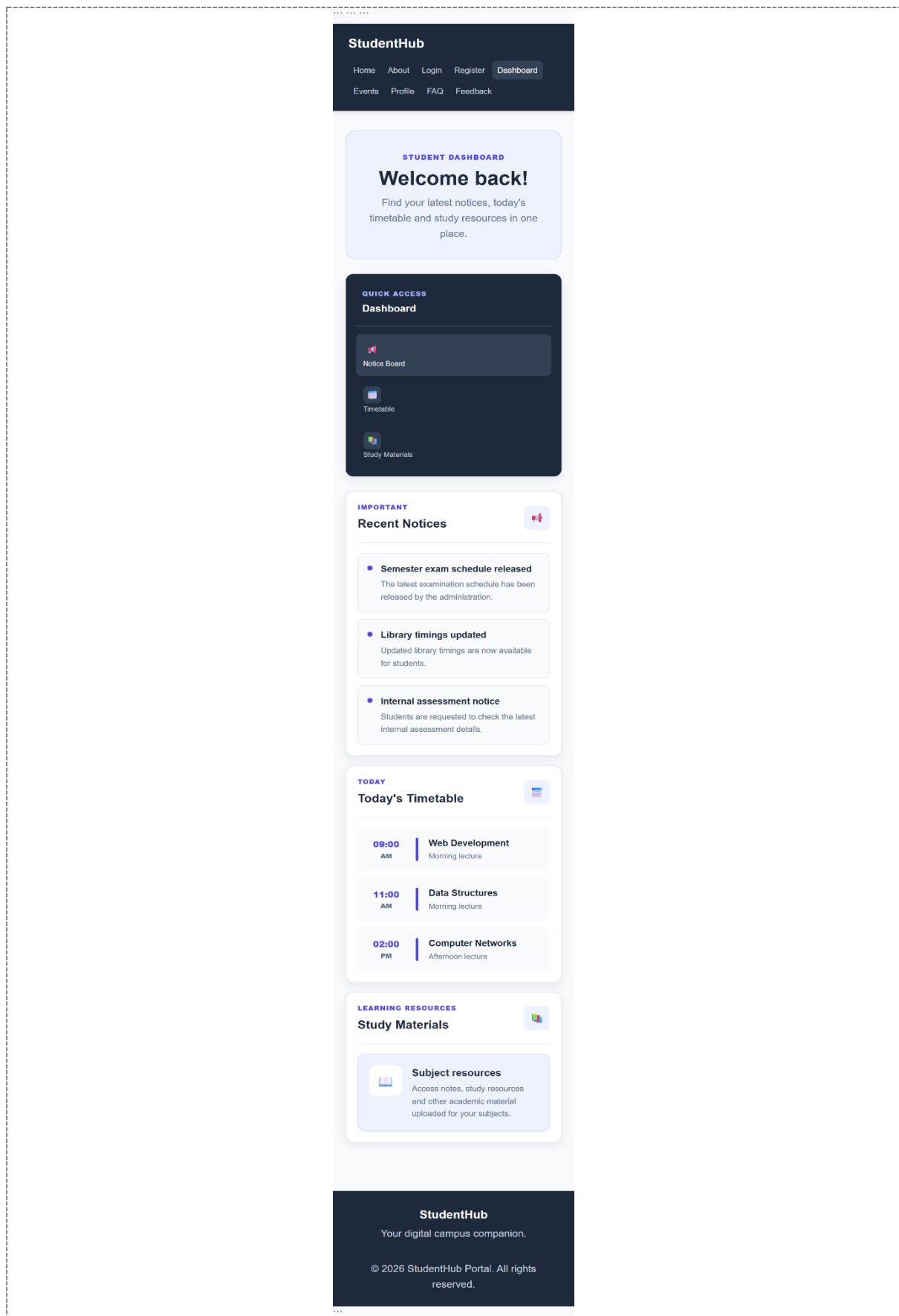


Figure: Dashboard page, mobile width

Events Page

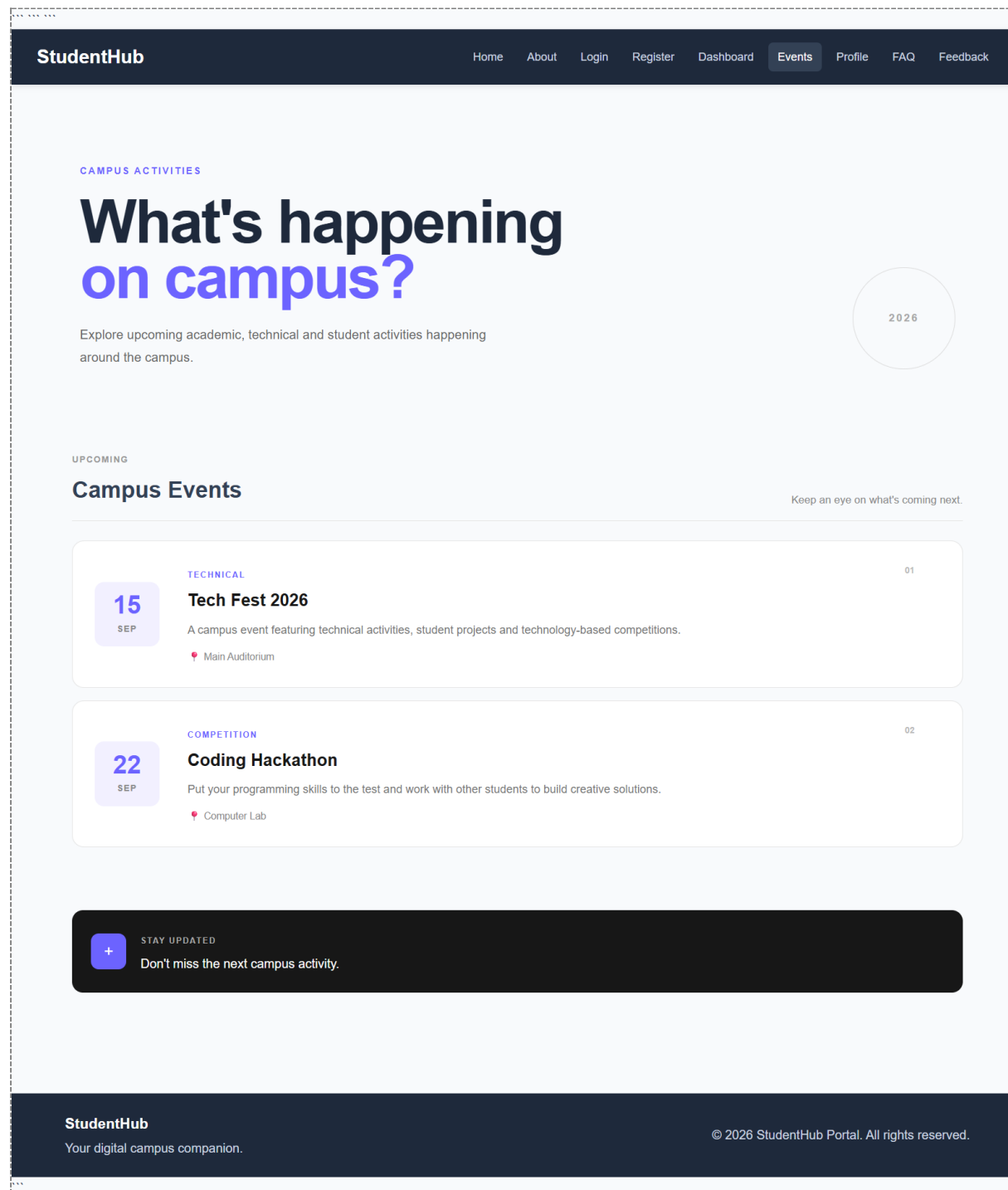


Figure: Events page, desktop width

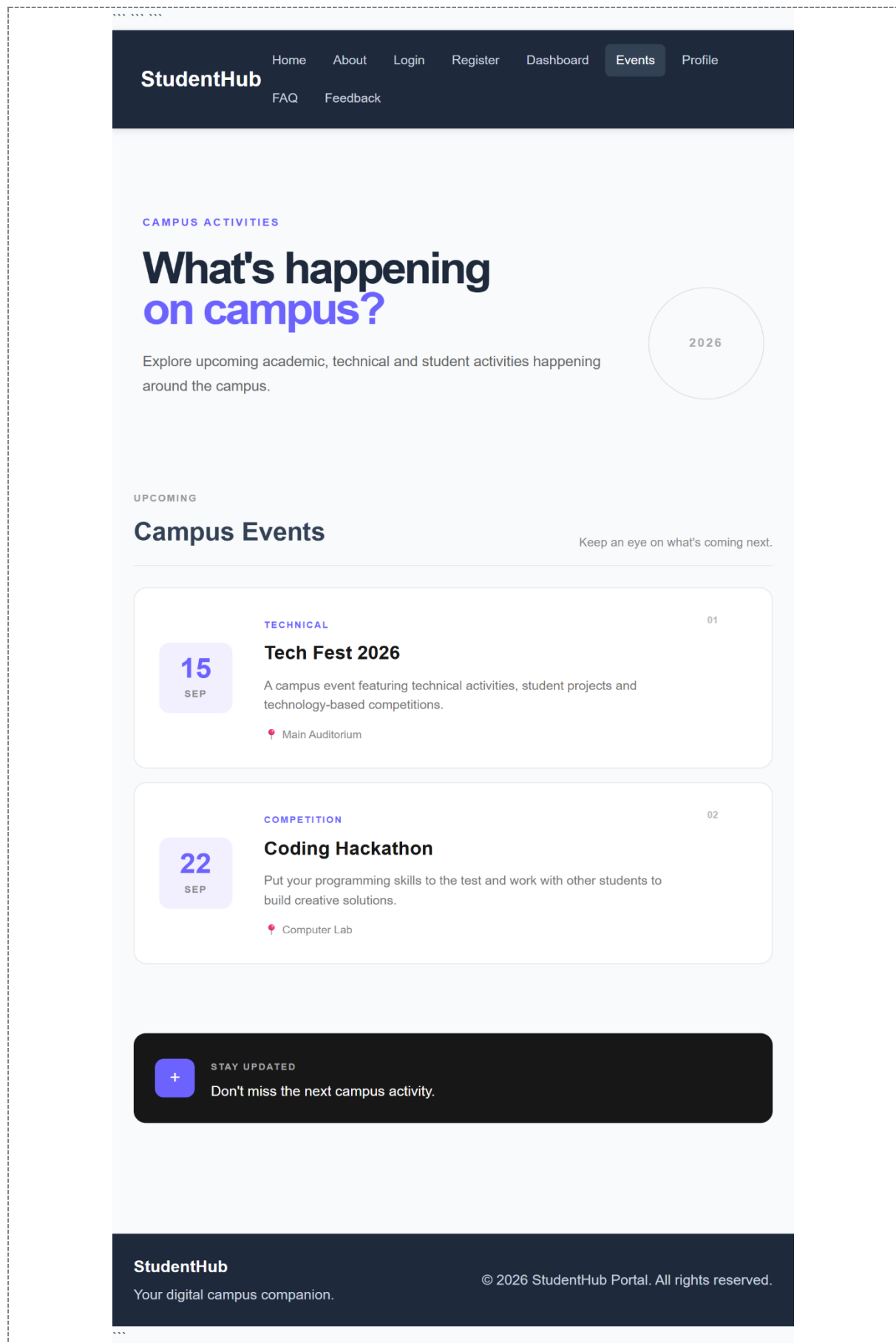


Figure: Events page, tablet width

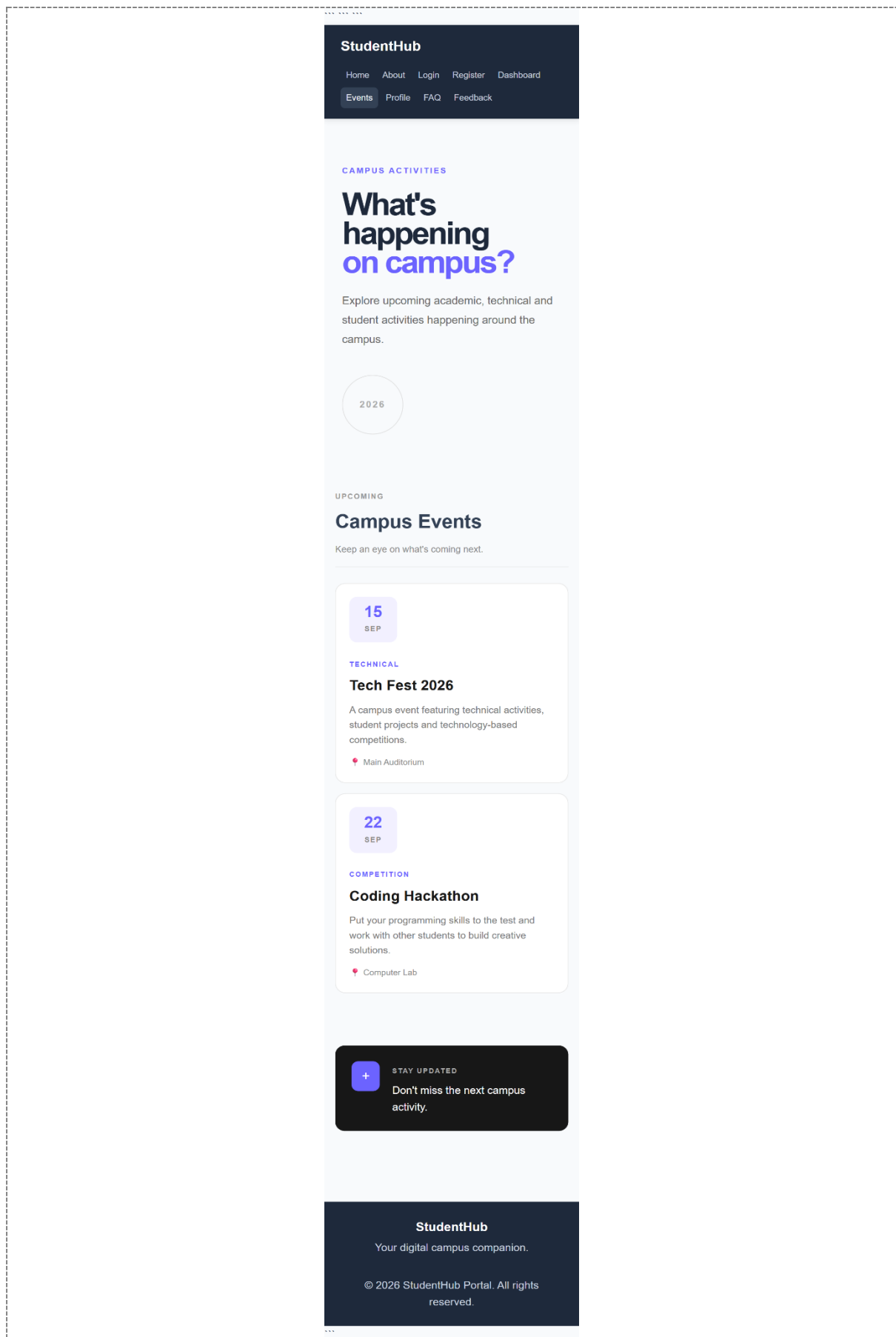


Figure: Events page, mobile width

Key Questions and Answers

Q1. How does the layout adapt for mobile, tablet and desktop screens?

The base CSS (outside any media query) targets mobile screens with single-column grids and stacked flex items. A media query at min-width: 768px changes selected grid-template-columns values to add a second column or a sidebar, for example turning .features-grid into two columns and .dashboard-layout into a 220px sidebar next to the content. A further media query at min-width: 1024px adds a third column to the features grid for desktop. Two max-width queries, at 900px and 767px, simplify layouts such as the registration form back to one column and adjust spacing once the screen is no longer wide enough for the tablet layout.

Q2. Which layout technique is used, and why?

CSS Grid is used for page-level sections that need rows and columns together, such as the feature cards, the about layout, the dashboard sidebar and the events list, because Grid can control both dimensions at once and change the column count with a single property. Flexbox is used for smaller, one-dimensional groupings, such as arranging the icon, heading, text and link inside a single feature card, and lining up the logo and navigation links in the header, because Flexbox handles alignment along one axis and vertical centring more simply than Grid.

Q3. Are typography, spacing, colours and contrast readable?

Font sizes and spacing use rem units, so they scale with the rest of the page instead of staying fixed in pixels. Padding increases at the tablet breakpoint so that sections do not look cramped on larger screens. Text and background colours are taken from the same CSS variables used across every StudentHub page, which keeps contrast consistent between pages and screen sizes.

Q4. Is the framework used consistently, without unnecessary inline CSS?

No CSS framework such as Bootstrap or Tailwind was used; the responsive behaviour was built directly with CSS Grid, Flexbox and media queries. All layout and spacing rules are kept in css/style.css, organised into numbered sections by page or component. No inline style attributes are used in the HTML files, which keeps the layout rules in one place and easy to update.

Post Laboratory Work:

Screenshots of the mobile, tablet and desktop views were captured for each page and are included above.

GITHUB REPOSITORY LINK:

<https://github.com/DarshilRupani/studenthub-portal>

(FOR SOURCE CODE)

Conclusion:

CSS Grid and Flexbox were used to make the Home, About, Registration, Dashboard and Events pages responsive across mobile, tablet and desktop widths, following a mobile-first approach with min-width and max-width media queries. Typography and spacing were kept consistent using rem units and shared CSS variables, and no inline styles or CSS framework were needed. Contact and Feedback pages were added as the intermediate extension, using the same responsive patterns as the rest of the site.